AI and Quantum Technologies

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CircleU Frontiers in Artificial Intelligence, April 15, 2026

Outline

The Big Picture

AI for Quantum Technologies

Quantum Computing for AI

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A Transformative Moment in Computational Science

Two revolutions are unfolding in parallel:

- **Artificial Intelligence** has transformed data-driven modelling, optimisation, and prediction
- **Quantum Technologies** promise fundamentally new approaches to computing, simulation, sensing, and communication

These two fields are now beginning to influence each other in profound ways.

Three Core Components

1. High-performance computing (HPC)
2. Artificial intelligence & ML
3. Quantum technologies

Together these may form a **new paradigm** for scientific discovery.

The Role of Computational Science Today

- Computational science is a central pillar of modern research, complementing theory *and* experiment
- Rapid growth of **HPC** enables increasingly realistic simulations across climate science, neuroscience, materials research, and astrophysics
- **AI and ML** have transformed how researchers analyse large datasets, discover patterns, and build predictive models
- **Quantum technologies** are now translating fundamental quantum-mechanical concepts — superposition, entanglement — into practical technologies

Quantum computers open new capabilities in simulation, optimisation, and linear algebra.

Machine learning meets quantum hardware

Core challenge: Quantum devices are delicate systems. Small imperfections $\Rightarrow$ decoherence and loss of information.

Reinforcement learning has proven particularly powerful:

- Agent interacts with a simulated/experimental quantum system
- Learns control protocols that optimise a desired objective
- Applications: state preparation, high-fidelity gate implementation

ML Applications to Quantum Hardware

- Calibration & characterisation
- Error mitigation & noise reduction
- Automated control-sequence discovery
- Adaptive quantum experiment design

Parameter spaces in quantum devices are **huge** — traditional optimisation struggles.

Can quantum hardware accelerate machine learning?

- Many ML algorithms rely on expensive **linear algebra**: matrix inversion, eigenvalue decomposition, high-dimensional optimisation
- Quantum algorithms have been proposed that may *accelerate* certain instances of these tasks: linear systems, sampling, optimisation
- The field of **quantum machine learning** explores entirely new computational models enabled by quantum hardware

Hybrid Quantum–Classical Algorithms

- A classical optimisation loop interacts with a quantum processor evaluating parameterised quantum circuits
- Key framework for *near-term* (NISQ) quantum devices
- Practical quantum advantage for ML remains an **open question**

Toward a New Paradigm for Computational Discovery

Three pillars — distinct but interacting:

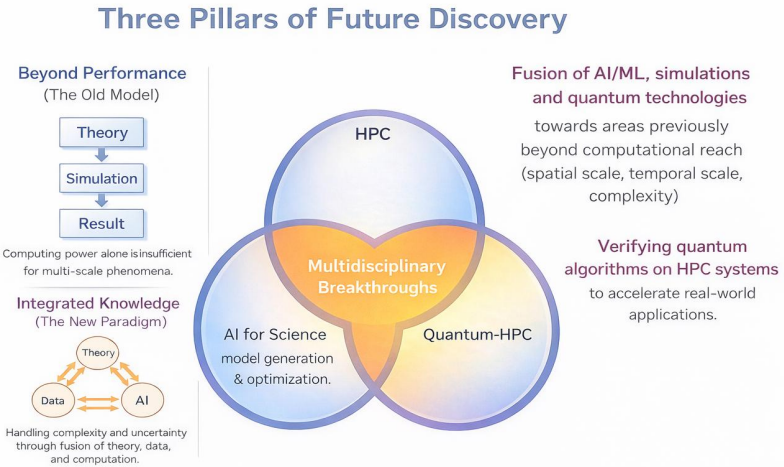
1. **High-performance computing**
   Large-scale simulations, data generation
2. **AI and machine learning**
   Pattern recognition, surrogate modelling, model reduction
3. **Quantum technologies**
   New capabilities for specific hard problems

Key implications:

- Experiments generate *massive datasets* requiring sophisticated analysis
- ML can *accelerate* simulations via surrogate models and effective physical representations
- Quantum technologies may enable simulations of *strongly correlated* systems intractable classically

Computational science becomes a tightly integrated ecosystem: theory, simulation, data analysis, and experiment interact continuously.

Conceptual Framework — Three Pillars of Future Discovery



Case Studies — Overview

Quantum Control

RL and neural-network optimisation discover control strategies that *outperform* traditional approaches for quantum computers and sensors

Quantum Simulation of Many-Body Systems

Classical approaches face *exponential* complexity for strongly correlated systems; hybrid quantum-classical methods may provide new tools

Quantum ML for Finance

- Portfolio optimisation via QAOA / quantum annealing
- Quantum kernel methods for financial time series
- Quantum generative models for risk analysis and Monte-Carlo-style simulations

Applications to Neuroscience

- QML for high-dimensional neural data representation

Spotlight: Quantum Control with Reinforcement Learning

Why is control hard?

- Parameter spaces of quantum devices are extremely large
- Traditional optimisation methods scale poorly
- Noise and decoherence degrade any imperfect protocol

What RL offers:

- Model-free exploration of huge control landscapes
- Discovers non-intuitive protocols
- Adapts online to hardware drift

Key References

- Bukov & Marquardt, *RL for Quantum Technology*, arXiv:2601.18953 (2026)
- Belliardo et al., *Model-aware RL for Bayesian quantum metrology*, Quantum **8**, 1555 (2024)

Result: RL agents have demonstrated gate fidelities competitive with hand-tuned protocols on real hardware.

Spotlight: Quantum ML for Finance and Neuroscience

Finance

- Large datasets, stochastic dynamics, high-dimensional optimisation
- **QAOA** / quantum annealing for combinatorial portfolio optimisation
- **Quantum kernel methods** — data encoded in quantum states; similarity evaluated in Hilbert space
- **Quantum generative models** — variational circuits or quantum Boltzmann machines for risk analysis

Neuroscience

- Brain: billions of interacting neurons, highly complex dynamical system
- ML already central for analysing neural recordings
- **QML** may represent high-dimensional neural data or model complex probability distributions
- Models of collective neural behaviour resemble **spin systems** from statistical physics $\Rightarrow$ methods from quantum many-body physics may offer new insights

Future Strategies and Requirements

Interdisciplinary Collaboration

Physics · Computer science · Mathematics · Engineering

Key challenges lie *at the intersection* of these fields

Education & Training

Future researchers need expertise in:

Physics · Numerical methods · Machine learning · Quantum information science

Research Infrastructure

Integrated platforms combining:

HPC resources · Experimental quantum platforms · ML-driven analysis environments

Realising this potential requires coordinated efforts across multiple disciplines — no single community can do it alone.

Reinforcement Learning and Quantum Control: Motivation

Many problems in physics can be phrased as optimization problems:

- variational calculations,
- control of dynamical systems,
- adaptive measurements,
- maximizing sensitivity in quantum sensors.

However, in many realistic settings:

- the objective is not known analytically,
- gradients are unavailable or noisy,
- decisions must be taken sequentially,
- present choices affect future opportunities.

Reinforcement learning (RL) addresses precisely this type of problem.

What is Reinforcement Learning? In plain words

The RL framework consists of an **agent** that learns how to solve a task by interacting with a physical system, called the environment. The agent can select **actions** that modify the **state** of the environment, as a consequence a **reward** is given to the agent, and the process repeats in a positive feedback loop. The objective is to maximise the sum of expected rewards andf thereby accomplish the task. The agent learns a **strategy** that can be meployed under similar yet different circumstances. The so-called **policy** os the rule for choosing actions.

My more narrow motivation (another talk perhaps)

- Quantum sensors permit fine-grained control over probe evolution and measurement settings.
- The central design problem is sequential: which control should be applied *next*, given what has already been learned?
- For realistic platforms, the estimation stack contains many non-smooth components: sampling of measurement outcomes, particle-filter resampling, and optional weak-measurement backaction.
- The article by Belliardo *et al.* (UniPI, SNS) proposes a model-aware RL framework that treats experimental design as a trainable control problem while retaining a faithful physical model.

What is Reinforcement Learning?

An agent interacts with an environment over discrete time steps:

$$s_0, a_0, r_1, s_1, a_1, r_2, \ldots$$

Core ingredients:

- **State** s_t: information available at time t,
- **Action** a_t: decision taken by the agent,
- **Reward** r_{t+1}: feedback from the environment,
- **Policy** $\pi(a|s)$: rule for choosing actions.

Goal:

$$\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t r_{t+1}\right]$$

where $0 \leq \gamma < 1$ is a discount factor.

Sequential Decision Making

In ordinary optimization we choose parameters once.
In reinforcement learning we choose a *sequence* of actions:

$$a_0, a_1, a_2, \ldots$$

This matters because:

- actions change the future state,
- the future state changes which actions are useful later,
- rewards may be delayed.

Thus RL is best viewed as **optimization over trajectories or strategies**.

Model-aware RL

Core idea

Use a differentiable simulation of the experiment, Bayesian inference by particle filtering, and policy optimization to learn adaptive or non-adaptive control laws for quantum metrology.

- **Bayesian mode:** optimize the final estimation error after a complete measurement sequence.
- **Frequentist mode:** optimize the Cramer-Rao bound through the Fisher information matrix.
- **Model-aware:** gradients are propagated through a known sensor model rather than inferred only from trial-and-error interaction.

The measurement loop as a control architecture

1. From the current posterior summary, the agent outputs the next control x_{t+1}.
2. The probe evolves and is measured, producing a stochastic outcome y_{t+1}.
3. The Bayesian posterior is updated and fed back to the agent.

Adaptive policies are written as

$$x_{t+1} = F_\lambda\big(P(\theta \mid x_t, y_t);\, y_t;\, R_t;\, t\big),$$

while non-adaptive strategies suppress dependence on the posterior and outcomes:

$$x_{t+1} = F_\lambda(R_t, t).$$

- λ: trainable parameters of the policy (typically a neural network).
- R_t: resources consumed up to step t (time, photons, signal amplitude, ...).

Model-Free vs Model-Aware Reinforcement Learning

Two broad philosophies:

Model-free RL

- learn directly from transitions,
- no explicit model of $P(s'|s,a)$ is used,
- examples: Q-learning, REINFORCE.

Model-aware (or model-based) RL

- use or learn a model of the dynamics,
- plan using that model,
- combine prediction and control.

In physics, model-aware RL is often especially attractive because we frequently know at least part of the dynamics.

What is Model-Aware Reinforcement Learning?

In model-aware RL we assume access to:

- known transition probabilities $P(s'|s,a)$, or
- an approximate learned model $\hat{P}_\phi(s'|s,a)$.

Then we can:

- simulate hypothetical futures,
- estimate returns before acting,
- plan over multiple time steps,
- use differentiable dynamics when available.

This is close in spirit to standard computational physics:

$$\text{equations of motion} \Rightarrow \text{simulation} \Rightarrow \text{optimization}.$$

Benefits of Model-Aware RL

Advantages:

- better sample efficiency,
- ability to plan ahead,
- easier incorporation of prior physical knowledge,
- natural interface with control theory.

Disadvantages:

- model bias if the model is wrong,
- errors may accumulate during planning,
- learning an accurate model may be difficult.

Thus model-aware RL is especially useful when the dynamics are fairly well understood, but the control task remains hard.

Model-Aware RL and Physics

In many physics applications we know the dynamics:

$$\dot{x} = f(x, u)$$

or in quantum mechanics

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = H(u(t))|\psi(t)\rangle.$$

Then the environment is not a black box:

- we can simulate trajectories,
- estimate gradients,
- evaluate candidate policies offline.

Model-aware RL thus sits naturally between:

- optimal control,
- stochastic optimization,
- machine learning.

Transition to Quantum Systems

Quantum control problems are naturally sequential:

- apply pulse,
- observe state evolution,
- choose next pulse,
- optimize final performance.

This makes RL conceptually attractive. At the same time, quantum dynamics are often known from the Schrödinger equation, which makes model-aware RL particularly relevant.

Quantum Control

Quantum dynamics:

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = H(t)|\psi(t)\rangle.$$

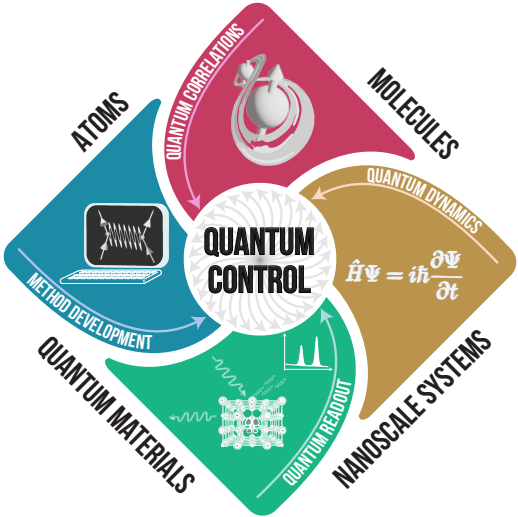
Controlled Hamiltonian:

$$H(t) = H_0 + \sum_k u_k(t) H_k$$

Typical goals:

- state preparation,
- gate synthesis,
- suppression of decoherence,
- enhancement of sensor performance.

Quantum Control



RL Formulation for Quantum Control

Mapping:

- **State**: wavefunction, density matrix, or measurement record,
- **Action**: control amplitude or pulse choice,
- **Reward**: fidelity, energy, Fisher information, robustness.

Examples of terminal rewards:

$$R = |\langle \psi_{\text{target}} | \psi(T) \rangle|^2$$

or

$$R = F_Q(T)$$

for a quantum sensing task.

Quantum Sensors and Fisher Information

A quantum sensor estimates a parameter θ.
Precision is bounded by the quantum Cramér–Rao bound:

$$\Delta\theta \geq \frac{1}{\sqrt{\nu F_Q}}$$

where:

- F_Q is the quantum Fisher information,
- ν is the number of repetitions.

Thus maximizing F_Q is a natural control objective for reinforcement learning.

RL Implementation Blueprint I

An actual implementation would follow these steps:

- **Define the environment:** Choose the sensor platform and identify its controllable parameters and observables. Build a simulation of the sensor dynamics (e.g. master equation or Monte Carlo) that includes the unknown parameter. If possible, validate the simulation against experiments.
- **State representation:** Decide what information to feed to the agent. This may be raw data (fluorescence images, spin readouts) or pre-processed features (means and variances). Include memory of previous steps if needed (e.g. via recurrent networks) to handle non-Markovian effects.

RL Implementation Blueprint II

- **Action space:** Specify the set of actions. For continuous controls, discretization or parameterized actions can be used. Ensure actions respect experimental limits (e.g. maximum pulse amplitude).
- **Reward function:** Choose a reward that captures precision. A terminal reward $R = -\log(\text{estimation variance})$ is one option. Alternatively, use stepwise rewards like Bayesian information gain. Regularize the reward if necessary to avoid trivial strategies.
- **RL algorithm:** Select an algorithm based on action type. For example, set neural network architectures appropriate for the state (e.g. convolutional nets for images).

RL Implementation Blueprint III

- **Training:** Train the agent in simulation. Introduce variability (randomize unknowns) to encourage generalization. Use experience replay or batch updates to improve sample efficiency. Monitor performance on validation scenarios.
- **Deployment:** Transfer the trained policy to the real device. If training was in-silico, consider fine-tuning on the hardware with a few adjustment episodes. Include safeguards for safety (e.g. bounding pulses) and mechanisms to reset the environment.

Worked Quantum Example: Two-Level Sensor

Consider a qubit sensing a magnetic field B:

$$H(t) = \frac{\omega_0}{2}\sigma_z + \frac{B}{2}\sigma_z + u_x(t)\sigma_x.$$

Interpretation:

- ω_0: bare splitting,
- B: unknown signal,
- $u_x(t)$: control field chosen by the agent.

Goal:

- accumulate phase information about B,
- keep coherence as long as possible,
- maximize final sensitivity.

Quantum Sensor as an RL Problem

Discretize time into steps $t_1, \ldots, t_N$.
Possible actions:

$$a_t \in \{-u, 0, +u\}.$$

Possible state representation:

- Bloch vector $(\langle\sigma_x\rangle, \langle\sigma_y\rangle, \langle\sigma_z\rangle)$,
- time step,
- previous measurement outcomes.

Possible reward:

$$R = F_Q(T)$$

or

$$R = \left| \frac{\partial}{\partial B} \langle\sigma_y(T)\rangle \right|.$$

Why RL for Quantum Sensors?

RL can learn how to:

- alternate between free precession and control,
- suppress unwanted evolution,
- balance phase accumulation against decoherence,
- adapt measurements based on partial information.

The learned strategy may resemble:

- Ramsey interferometry,
- spin echo,
- dynamical decoupling,
- adaptive phase estimation.

Model-Aware RL for Quantum Control

Quantum systems are a prime example of model-aware RL.
If the Hamiltonian is known, we can:

- simulate trajectories offline,
- evaluate hypothetical pulse sequences,
- use differentiable propagators,
- combine RL with standard optimal-control ideas.

This is especially attractive for:

- quantum control,
- calibration,
- sensor design,
- adaptive metrology.

RL and Quantum Control Summary

Reinforcement learning provides:

- a framework for sequential decision making,
- value-based and policy-based algorithms,
- a natural extension of stochastic optimization,
- direct relevance for quantum control and quantum sensing.

For physicists, the key conceptual bridge is:

$$\text{variational optimization} + \text{stochastic dynamics} + \text{control} \;\Rightarrow\; \text{reinforcement learning}.$$

Conclusions

- AI and quantum technologies are **rapidly reshaping** the landscape of computational science
- Their mutual interaction, combined with advances in HPC, is transforming how scientific problems are approached and solved
- The integration of **theory, simulation, machine learning, and quantum hardware** may enable entirely new forms of scientific discovery
- Promising near-term applications span: quantum control, many-body simulation, hardware development, finance, and neuroscience

As these technologies continue to mature, their convergence will play a central role in addressing some of the most challenging problems in physics, chemistry, and materials science.

Key References

- I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press (2016).
- J. Preskill, Quantum computing in the NISQ era, *Quantum* **2**, 79 (2018).
- M. Bukov and F. Marquardt, Reinforcement learning for Quantum Technology, arXiv:2601.18953 (2026).
- F. Belliardo, F. Zoratti, F. Marquardt, and V. Giovannetti, Model-aware reinforcement learning for high-performance Bayesian experimental design in quantum metrology, *Quantum* **8**, 1555 (2024).
- M. Schuld and F. Petruccione, *Machine Learning with Quantum Computers*, Springer-Verlag (2021).

Additional material on RL

Markov Decision Process

The standard framework is a Markov Decision Process (MDP).
Elements:

- state space $\mathcal{S}$,
- action space $\mathcal{A}$,
- transition probability $P(s'|s,a)$,
- reward function $R(s,a)$ or $R(s,a,s')$.

Markov property:

$$P(s_{t+1}|s_t,a_t,s_{t-1},a_{t-1},\dots) = P(s_{t+1}|s_t,a_t)$$

This is analogous to stochastic evolution equations in statistical physics.

Return and Value Functions

Define the return from time t:

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$

The **state-value function** under policy π is

$$V^\pi(s) = \mathbb{E}_\pi[G_t \mid s_t = s]$$

The **action-value function** is

$$Q^\pi(s, a) = \mathbb{E}_\pi[G_t \mid s_t = s,\ a_t = a]$$

The **advantage** is

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$$

Interpretation: how much better is action a than the typical action at state s?

Bellman Equation

The value function satisfies a recursive relation:

$$V^\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s,a) \left[r(s,a,s') + \gamma V^\pi(s') \right]$$

This is the Bellman equation.
Interpretation:

$$\text{value} = \text{immediate reward} + \text{discounted future value}$$

This is a self-consistency relation, reminiscent of mean-field equations and iterative schemes in physics.

Optimal Bellman Equation

Define the optimal value

$$V^*(s) = \max_\pi V^\pi(s)$$

and optimal action-value function

$$Q^*(s,a) = \max_\pi Q^\pi(s,a).$$

Then

$$V^*(s) = \max_a \sum_{s'} P(s'|s,a)\left[r(s,a,s') + \gamma V^*(s')\right]$$

and

$$Q^*(s,a) = \sum_{s'} P(s'|s,a)\left[r(s,a,s') + \gamma \max_{a'} Q^*(s',a')\right].$$

This is the basis for dynamic programming and Q-learning.

Value Iteration and Policy Iteration

Value iteration:

$$V_{k+1}(s) = \max_a \sum_{s'} P(s'|s,a)\left[r + \gamma V_k(s')\right]$$

Policy iteration:

1. Evaluate V^π for current policy,
2. Improve the policy using

$$\pi_{\text{new}}(s) = \arg\max_a Q^\pi(s,a)$$

These are exact if the model is known and the state-action spaces are manageable.

Q-Learning

When the model is unknown, we can learn directly from sampled transitions.
Q-learning update:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right]$$

Here:

- α is the learning rate,
- the term in brackets is the temporal-difference error.

This is a stochastic approximation to the Bellman optimality equation.

Exploration vs Exploitation

A central issue in RL is the exploration–exploitation tradeoff.

- **Exploitation**: choose the best-known action,
- **Exploration**: try other actions to discover better strategies.

Common strategies:

- ε-greedy,
- softmax/Boltzmann exploration,
- entropy regularization.

This is analogous to balancing local minimization and global search in complex optimization landscapes.

Why Policy Gradient Methods?

Q-learning is natural for discrete action spaces.
But in physics and control problems we often have:

- continuous control amplitudes,
- stochastic policies,
- large state spaces,
- differentiable parameterizations.

Then it is natural to parameterize the policy directly:

$$\pi_\theta(a|s)$$

and optimize the expected return

$$J(\theta).$$

Policy Gradient Setup

Define

$$J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}[R(\tau)]$$

where $\tau = (s_0, a_0, s_1, a_1, \dots)$ is a trajectory and

$$R(\tau) = \sum_{t=0}^{\infty} \gamma^t r_{t+1}.$$

Trajectory probability:

$$P_\theta(\tau) = \rho(s_0) \prod_{t=0}^{\infty} \pi_\theta(a_t|s_t) P(s_{t+1}|s_t, a_t)$$

We want to compute

$$\nabla_\theta J(\theta).$$

Policy Gradient Theorem: Step 1

Start with

$$J(\theta) = \sum_{\tau} P_\theta(\tau) R(\tau).$$

Differentiate:

$$\nabla_\theta J(\theta) = \sum_{\tau} \nabla_\theta P_\theta(\tau)\, R(\tau).$$

Use the log-derivative identity:

$$\nabla_\theta P_\theta(\tau) = P_\theta(\tau) \nabla_\theta \log P_\theta(\tau).$$

Hence

$$\nabla_\theta J(\theta) = \sum_{\tau} P_\theta(\tau) \nabla_\theta \log P_\theta(\tau)\, R(\tau).$$

Therefore

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\nabla_\theta \log P_\theta(\tau)\, R(\tau) \right].$$

Policy Gradient Theorem: Step 2

Now expand the log trajectory probability:

$$\log P_\theta(\tau) = \log \rho(s_0) + \sum_{t=0}^{\infty} \log \pi_\theta(a_t|s_t) + \sum_{t=0}^{\infty} \log P(s_{t+1}|s_t, a_t).$$

Only the policy depends on θ, so

$$\nabla_\theta \log P_\theta(\tau) = \sum_{t=0}^{\infty} \nabla_\theta \log \pi_\theta(a_t|s_t).$$

Substituting,

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\left(\sum_{t=0}^{\infty} \nabla_\theta \log \pi_\theta(a_t|s_t)\right) R(\tau)\right].$$

Policy Gradient Theorem: Step 3

We can refine this by assigning to each time step only the future return from that point onward:

$$G_t = \sum_{k=t}^{\infty} \gamma^{k-t} r_{k+1}.$$

Then

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{\infty} \nabla_\theta \log \pi_\theta(a_t|s_t)\, G_t\right].$$

This is the basis of the REINFORCE algorithm.
A more compact form is

$$\nabla_\theta J(\theta) = \mathbb{E}_{s\sim d^\pi,\, a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s)\, Q^{\pi_\theta}(s,a)\right]$$

where $d^\pi(s)$ is the discounted state visitation distribution.

Variance Reduction and Baselines

The estimator can have high variance.
We may subtract a baseline $b(s)$ without changing the expectation:

$$\nabla_\theta J(\theta) = \mathbb{E}\left[\nabla_\theta \log \pi_\theta(a|s)\,(Q^\pi(s,a) - b(s))\right].$$

A natural choice is

$$b(s) = V^\pi(s),$$

which gives

$$\nabla_\theta J(\theta) = \mathbb{E}\left[\nabla_\theta \log \pi_\theta(a|s)\,A^\pi(s,a)\right].$$

This motivates actor–critic methods.

REINFORCE Algorithm

Basic Monte Carlo policy gradient algorithm

For each episode:

1. Generate a trajectory using π_θ,
2. Compute returns G_t,
3. Update parameters:

$$\theta \leftarrow \theta + \alpha \sum_t \nabla_\theta \log \pi_\theta(a_t|s_t)\, G_t$$

Advantages:

- simple,
- direct,
- works with stochastic policies.

Disadvantage:

- large variance.

Actor–Critic Methods

Actor–critic combines:

- **Actor**: policy $\pi_\theta(a|s)$,
- **Critic**: estimate of value function $V_\phi(s)$.

Temporal-difference error:

$$\delta_t = r_{t+1} + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$$

Policy update:

$$\theta \leftarrow \theta + \alpha_\theta \nabla_\theta \log \pi_\theta(a_t|s_t)\, \delta_t$$

Critic update:

$$\phi \leftarrow \phi - \alpha_\phi \nabla_\phi \delta_t^2$$

This is often much more efficient than pure Monte Carlo policy gradients.

Worked RL Example: Grid World

Consider a particle moving on a 1D lattice:

$$x \in \{0, 1, 2, 3, 4\}.$$

Actions:

$$a \in \{\text{left}, \text{right}\}.$$

Goal:

- reach $x = 4$,
- avoid spending too many steps,
- possibly avoid a penalty region.

Reward example:

$$r = \begin{cases} +10, & x = 4, \\ -1, & \text{each intermediate step.} \end{cases}$$

This is simple enough to analyze explicitly, but already contains the core RL ideas.